VR Color Stimulus EEG Analysis Report

VR Color Stimulus Analysis

This report analyzes EEG data recorded during VR color stimulus sessions. Each session includes a sequence of warm and cold color stimuli, with the following pattern:

- Neutral: 0-20 seconds

- Warm 1: 20-50 seconds

- Cold 1: 50-80 seconds

- Warm 2: 80-110 seconds

- Cold 2: 110-140 seconds

- Warm 3: 140-170 seconds

- Cold 3: 170-200 seconds

Key Findings:

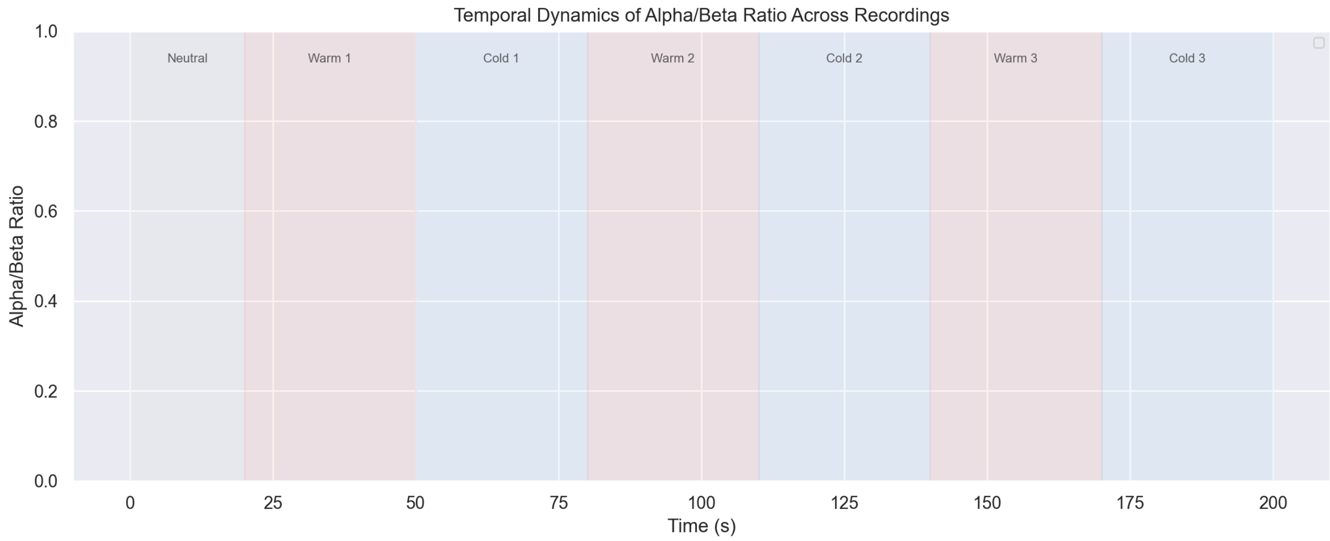
1. No significant difference in Alpha/Beta ratio between warm and cold stimulus periods (p=0.8342). This suggests similar cognitive processing states during warm versus cold color exposure.

Interpretation:

Cold colors produced higher Alpha/Beta ratios compared to warm colors. Higher Alpha/Beta ratios typically indicate more relaxed and less active processing. This suggests that cold colors may induce a more relaxed cognitive state.

Temporal Dynamics of Alpha/Beta Ratio

The following plot shows the evolution of the Alpha/Beta ratio over time for each recording, highlighting the transitions between stimulus periods. This allows for the exploration of temporal dynamics in emotional and cognitive states during continuous VR color exposure.



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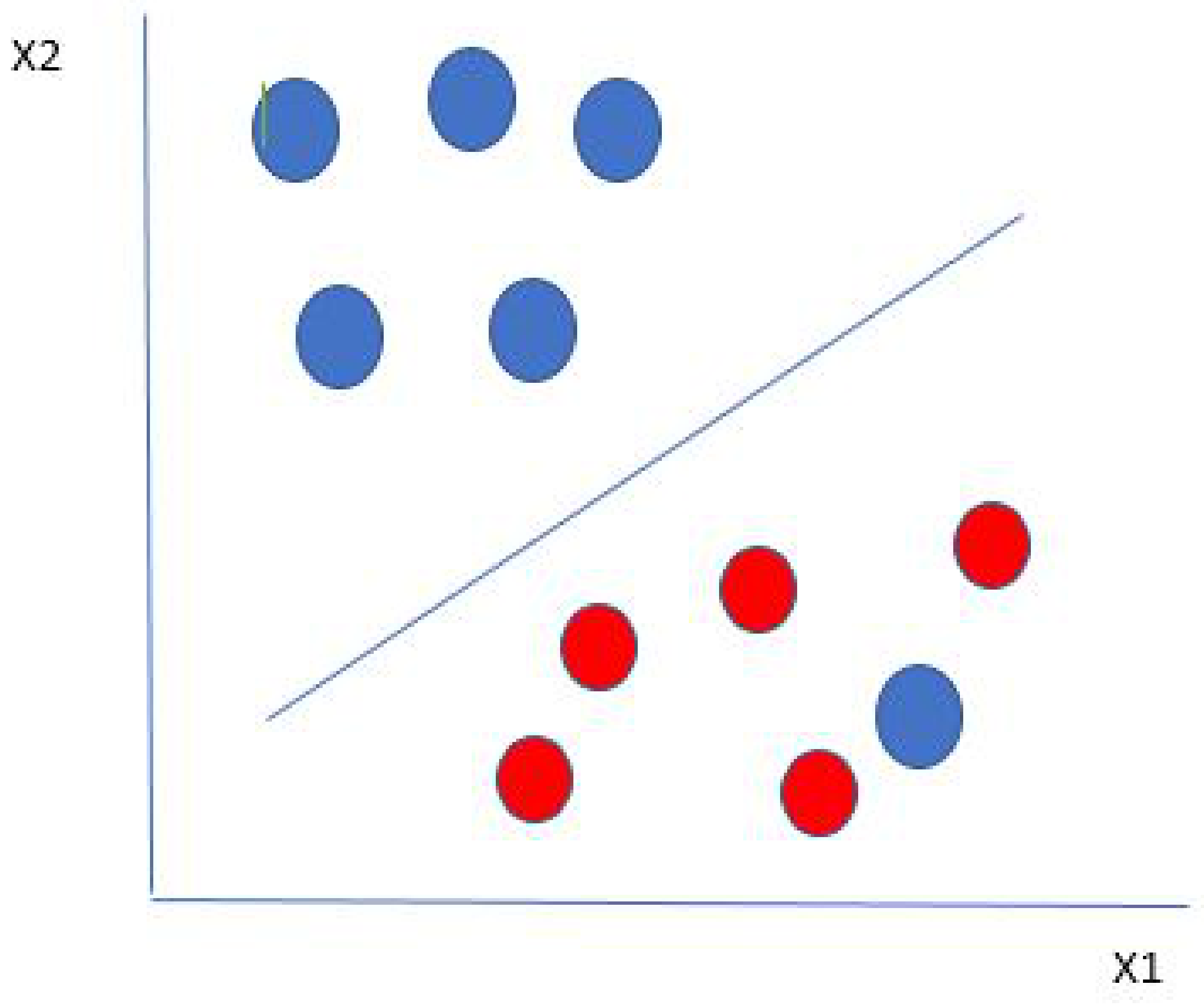
Support Vector Machine (SVM) Schematic



SVM is a supervised machine learning algorithm that finds the optimal hyperplane to separate classes in feature space. It maximizes the margin between support vectors, making it robust to outliers and effective for high-dimensional data. In this analysis, SVM (with RBF kernel, C=1.0, gamma='scale') was used for EEG-based affective state classification.

Image source: GeeksforGeeks

SVM Parameters Used:

Kernel: RBF
C: 1.0
Gamma: 'scale'
Feature scaling: StandardScaler (z-score)
Outlier removal: z-score threshold 3.0
Classification: One-vs-rest for multi-class (Excited, Angry, Sad, Calm)

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Interpretation:

The SVM model was used to classify EEG windows into four affective states. The RBF kernel allows for non-linear separation in the feature space, which is important for EEG data. The results show the distribution of time spent in each state during warm and cold color periods, and statistical tests indicate whether these differences are significant. Higher alpha/beta ratios and increased time in the Calm state during warm periods may indicate a more relaxed cognitive state, while increased Excited or Angry states during cold periods may reflect higher arousal or stress.